EE3032 Introduction to VLSI Design—Homework 3

1. Design a 3-bit ripple carry adder which consists of four full adders as shown in Fig. 1 using CMOS logic design style. Assume that every output of the adder drives a 1fF capacitance. Determine the transistor sizes arbitrarily of the 3-bit ripple carry adder and verify your design using Hspice simulator. Also, measure the worst-case propagation delay and the average power consumption of the ripple carry adder. (10 points)

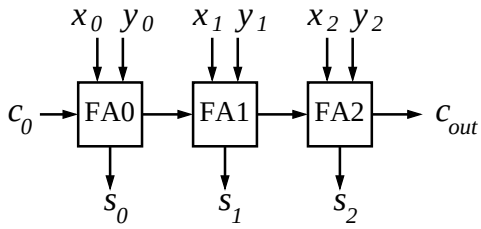

Figure 1: A 3-bit ripple carry adder.